

*Please ensure this tracking sheet is completed as a group, as it will be used to monitor your individual participation within the team.*

| Team Members                | Role/ Responsibility                                                                                                               | Email                           |            |                                                                              |                            |                                                                                          |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|------------|------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------|
| Belyse Intwaza              | Team Lead                                                                                                                          | b.intwaza@alustudent.com        |            |                                                                              |                            |                                                                                          |
| Saad Byiringiro             | Team Member                                                                                                                        | s.byiringir@alustudent.com      |            |                                                                              |                            |                                                                                          |
|                             |                                                                                                                                    |                                 |            |                                                                              |                            |                                                                                          |
|                             |                                                                                                                                    |                                 |            |                                                                              |                            |                                                                                          |
|                             |                                                                                                                                    |                                 |            |                                                                              |                            |                                                                                          |
| Task Allocation and Tracker |                                                                                                                                    |                                 |            |                                                                              |                            |                                                                                          |
|                             | Task Allocated                                                                                                                     | Assigned Member                 | Deadline   | Completion Status<br>(Not Started/In Progress/Completed/Did not participate) | Reviewed by Team? (Yes/No) | Comments                                                                                 |
|                             | Find an African dataset (NISR Rwanda Seasonal Agricultural Survey 2024 Season B), upload raw data and set up GitHub repo           | Belyse Intwaza                  | 30/09/2026 | Completed                                                                    | Yes                        | Raw CSV has missing values and text columns; metadata added; repo link added to notebook |
|                             | Data preparation: explore columns, handle missing values, encode text columns, save processed CSV                                  | Belyse Intwaza                  | 30/09/2026 | Completed                                                                    | Yes                        | PCA_Formative2_Person1.ipynb; 17,260 rows x 20 features, no missing values               |
|                             | Task 1: PCA from scratch (centering, covariance matrix, eigenvalues/eigenvectors, sorting, projection)                             | Belyse Intwaza                  | 30/09/2026 | Completed                                                                    | Yes                        | PCA_Formative2_Person1.ipynb; numpy + matplotlib only, checked against np.cov            |
|                             | Before/after PCA visualisations and interpretation of PC1 and PC2                                                                  | Belyse Intwaza                  | 30/09/2026 | Completed                                                                    | Yes                        | Covariance heatmap, before_pca.png and after_pca.png in figures/                         |
|                             | Task 2: explained variance, cumulative variance, dynamic selection of k (scree and cumulative plots, information lost per feature) | Saad Byiringiro                 | 30/09/2026 | Completed                                                                    | Yes                        | PCA_Formative2_Person2.ipynb; 90% threshold gives k = 15 (92.1% of variance kept)        |
|                             | Task 2: written explanation (why k, tradeoff, information lost)                                                                    | Saad Byiringiro                 | 30/09/2026 | Completed                                                                    | Yes                        | Markdown cell in PCA_Formative2_Person2.ipynb                                            |
|                             | Task 3: optimized chunked PCA, benchmark of time and memory, 50-million-row streaming test                                         | Saad Byiringiro                 | 30/09/2026 | Completed                                                                    | Yes                        | About 3x faster and 3.5x less memory at 3M rows; same results as the original            |
|                             | Task 3: written explanation of the improvements, and README update                                                                 | Saad Byiringiro                 | 30/09/2026 | Completed                                                                    | Yes                        | Markdown cell in PCA_Formative2_Person2.ipynb; README lists both notebooks               |
|                             | Final notebook review, combined PDF and submission                                                                                 | Belyse Intwaza, Saad Byiringiro | 30/09/2026 | Completed                                                                    | Yes                        | Both check before submitting                                                             |
|                             |                                                                                                                                    |                                 |            |                                                                              |                            |                                                                                          |
|                             |                                                                                                                                    |                                 |            |                                                                              |                            |                                                                                          |

| Meeting Attendance Log |                                                                            |             |                                 |                                                                                                                                       |                                                              |  |
|------------------------|----------------------------------------------------------------------------|-------------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|--|
| Meeting Date           | Agenda                                                                     | Facilitator | Attendees                       | Key Discussion Points                                                                                                                 | Next Steps                                                   |  |
| 25/09/2026             | Introductions, go through the assignment brief together and allocate tasks |             | Belyse Intwaza, Saad Byiringiro | Read the brief together and split the work: Belyse on dataset, data preparation and Task 1; Saad on Task 2, Task 3 and interpretation | Belyse to find an African dataset for the assignment         |  |
| 26/09/2026             | Check the dataset Belyse found                                             |             | Belyse Intwaza, Saad Byiringiro | Reviewed the NISR Rwanda Seasonal Agricultural Survey 2024 Season B data we will work with                                            | Belyse: data preparation and Task 1; Saad: Task 2 and Task 3 |  |
| 30/09/2026             | Final review and submission                                                |             | Belyse Intwaza, Saad Byiringiro | Final review of the notebooks, explanations and task sheet before submitting                                                          | Submit the assignment                                        |  |